
Project Proposal: Cooking Recipes Rating Based on Sentiment Analysis

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1. Project Title

Cooking Recipes Rating Based on Sentiment Analysis

An intelligent system designed to analyze user feedback and provide accurate recipe ratings, enabling users to discover trustworthy recipes efficiently.

2. Introduction

The main objective of this project is to enhance recipe discovery by analyzing user-generated feedback using Natural Language Processing (NLP) techniques, including Sentiment Analysis and Keyword Extraction. Unlike traditional numeric rating systems, this approach evaluates textual comments to provide meaningful insights and more accurate recipe recommendations.

This proposal is based on the SRS document, which outlines the system's functional and non-functional requirements, user roles, constraints, and technical dependencies. The system is intended for desktop web browsers on Windows, with all data stored in a centralized SQL database.

3. Objectives

1. Allow users to register, submit recipes, and provide feedback or comments.
2. Analyze user comments to determine sentiment (positive, neutral, or negative).
3. Extract key ingredients and cooking steps from comments to improve rating accuracy.
4. Identify potentially harmful ingredients and flag suspicious or fake reviews using Enhanced Random Forest (ERF).
5. Provide administrators with tools to manage users, recipes, comments, and keyword definitions.
6. Apply Documentation Techniques to ensure clarity, maintainability, and usability for developers, testers, and stakeholders.

Documentation Techniques Implementation:

- **Getting Started Guide:** Instructions to install and run the system.
 - **How-To Guides:** Step-by-step instructions for analyzing a recipe comment.
 - **Procedural Guide:** Full workflow from comment submission to sentiment rating.
 - **Code Comments:** Explain key code sections, including NLP and ERF algorithms.
 - **README:** Project overview, setup instructions, troubleshooting, and examples.
 - **Conceptual Documentation:** Explanation of NLP concepts, system architecture, and workflows.
 - **Reference Documentation:** API endpoints, database schema, and function descriptions.
 - **Glossary:** Definitions of technical terms such as Sentiment Analysis, ERF, NLP, BoW.
 - **Troubleshooting Documentation:** Common issues and solutions (e.g., database or translation API).
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4. Scope

In-Scope:

- Web-based platform for submitting, viewing, and rating recipes.
- Sentiment analysis and keyword extraction on user comments.
- Administrator interface for content moderation, keyword management, and flagged review handling.
- Desktop Windows environment with a centralized SQL Server database.

Out-of-Scope:

- Mobile platform support in the initial release.
 - Integration with external recipe APIs (except Google Translate API for multilingual comments).
 - Multimedia input (voice/video) and real-time review scraping in the first version.
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5. Methodology

The project will follow a Waterfall / structured development approach, ensuring clear steps, accurate outcomes, and proper documentation.

Step 1: Data Collection

- Gather user-submitted recipes, comments, and reviews.

- Ensure diverse and representative data to support analysis.

Step 2: Preprocessing

- Clean and normalize text by removing noise, punctuation, and irrelevant symbols.
- Translate non-English comments using Google Translate API.
- Prepare data in a structured format for analysis.

Step 3: Sentiment Analysis & Fake Review Detection

- Apply NLP techniques to classify comments as positive, neutral, or negative.
- Detect suspicious or fake reviews using Enhanced Random Forest (ERF).
- Ensure high accuracy and reliability in classification and review flagging.

Step 4: Backend & Database

- Store all data in Microsoft SQL Server, including users, recipes, comments, ratings, and keywords.
- Design an optimized centralized database for efficient data retrieval.
- Implement security measures to protect sensitive user data.

Step 5: Frontend & User Interface (UI)

- Develop a desktop web interface for both users and administrators.
- Include user-friendly features: recipe submission, comment posting, sentiment filtering, and admin monitoring tools.
- Ensure clear presentation and intuitive navigation.

Step 6: Testing & Validation

- Conduct unit, integration, and system testing.
- Validate sentiment analysis, keyword extraction, and fake review detection accuracy.
- Identify and correct errors in backend and frontend systems.

Step 7: Documentation Implementation

- Apply all documentation techniques for clarity and maintainability:
 - Getting Started Guide, How-To Guides, Procedural Guides
 - Code Comments, README, Conceptual Documentation
 - Reference Documentation, Glossary, Troubleshooting
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6. Expected Outcomes

- Fully functional web-based system for recipe rating and comment analysis.
 - Accurate sentiment-based recipe ratings.
 - Highlighted keywords and flagged suspicious reviews for administrative oversight.
 - Comprehensive documentation for developers and end-users.
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7. Benefits

- Enable users to quickly discover high-quality recipes.
 - Provide deeper insights than numeric ratings alone.
 - Increase trust in online recipes by detecting fake or unsafe content.
 - Enhance maintainability and collaboration through thorough documentation.
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8. Timeline (Gantt-style overview)

Phase	Duration	Deliverables
Requirements & SRS finalization	1 week	SRS document, use cases, glossary
System Design	1 week	Architecture diagrams, database schema
Backend Implementation	2 weeks	Database setup, sentiment analysis module
Frontend Implementation	2 weeks	User & admin UI, recipe submission, comment forms
Testing & Debugging	1 week	Unit, integration, system tests
Documentation	1 week	Guides, code comments, README, reference docs
Deployment & Presentation	1 week	Fully functional system, proposal/report submission

9. User Characteristics

- Regular Users: Register, submit recipes, leave comments, and view ratings.
 - Administrators: Moderate content, manage keywords and toxic ingredients, monitor platform activity.
 - Assumption: Users are comfortable using a web browser; no advanced technical skills required.
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10. Constraints & Assumptions

- Sentiment analysis accuracy depends on the quality of user comments.
 - Internet connection is required for submitting comments and using Google Translate API.
 - Desktop Windows browsers only; mobile devices not supported in this version.
 - SQL database performance may degrade under high user load; indexing required.
 - Google Translate API availability is assumed for multilingual support.
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11. Technical Dependencies

- **.NET Framework** for backend and UI.
 - **Microsoft SQL Server 2008** for data storage.
 - **Google Translate API** for non-English comment processing.
 - **Visual Studio 2010** for development.
 - NLP libraries: **NLTK**, **TextBlob** (optional Python integration).
 - **Enhanced Random Forest** algorithm for fake review detection.
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12. References

1. Text Preprocessing in NLP – Analytics Vidhya.
2. IEEE Paper on Sentiment Analysis.
3. Data Extraction Techniques – CData Blog